Lecture 2: Fatigue — Stress-Life and Strain-Life

UCK 364E Design of Machine Elements

Prof. Dr. Onur Tunçer

Fall 2026

Department of Aeronautical Engineering, Istanbul Technical University

Today

1. Why fatigue, not static strength, sizes an aircraft
2. The three stages of damage and how to read a fracture surface
3. Describing a cycle: σ_a, σ_m, R, $\Delta\sigma$
4. S–N curves, the endurance limit, and why aluminium has none
5. From specimen to component: the Marin factors
6. Notches: $K_t \to K_f$ and notch sensitivity
7. Mean stress: Goodman, Gerber, Soderberg, Morrow
8. Variable amplitude: Miner's rule and rainflow counting
9. Strain-life: Coffin–Manson and Neuber's rule
10. Safe-life, fail-safe, damage tolerance — and the scatter factor

Why fatigue governs

Last time every margin was huge

- Fuselage skin: $\mathrm{MS}_{\mathrm{ult}} = 2.86$
- Wing lug in bearing: $\mathrm{MS}_{\mathrm{ult}} = 0.70$
- Actuator rod end: $\mathrm{MS}_{\mathrm{y}} = 5.6$

Nothing on an aircraft is sized by *one* application of load. What sizes it is repetition far below the static allowable:

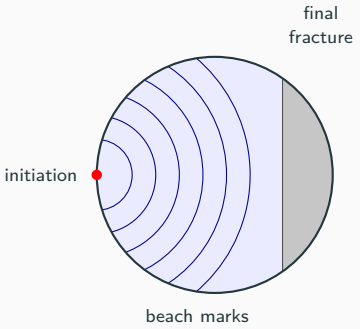
- 1 pressurisation cycle per flight
- 1 landing impact per flight
- gusts, every hour of cruise
- propeller / rotor vibration at 10–1000 Hz

The central fact

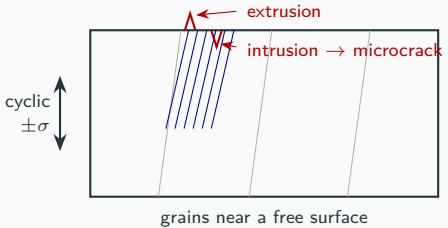
A metal loaded repeatedly well below yield will eventually crack. No static margin protects you. Only: lower cyclic stress, fewer cycles, or find the crack first.

Three stages of fatigue damage

1. Initiation — cyclic slip, intrusions and extrusions, a microcrack one grain across. At a *free surface*, at a *stress concentration*. Up to 90% of life in a polished specimen; almost none in a notched or corroded part.
2. Stable growth — one striation per cycle; beach marks record load changes. Described by fracture mechanics (next lecture).
3. Final fracture — the remaining section fails statically, in one cycle.

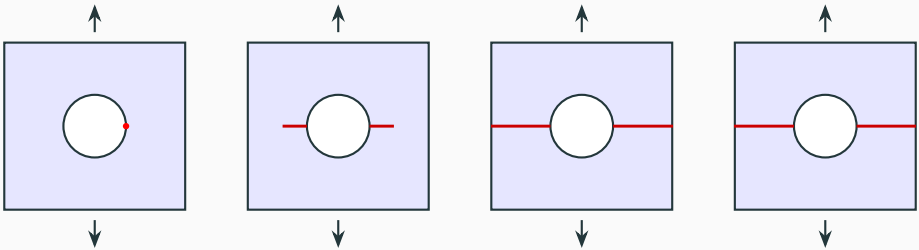


How a crack is born: slip, intrusions, extrusions



Cyclic slip in favourably oriented *surface* grains is irreversible: each cycle leaves a step. Steps pile up into an extrusion and its matching intrusion — a notch one grain deep that the next cycles turn into a microcrack. This is why fatigue is a surface phenomenon, and why surface finish, peening and corrosion matter more than anything else you can change.

The life of a crack at a fastener hole



(a) nucleation
$a \sim$ grain size

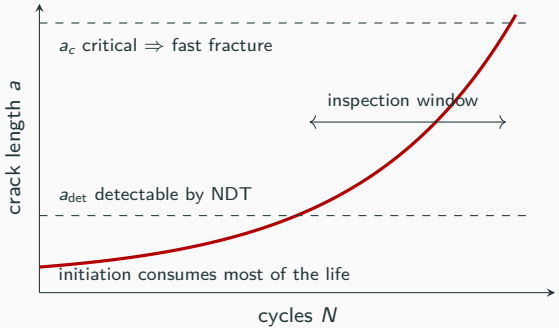
(b) short crack
$a < 1\,\text{mm}$

(c) long crack
growth, striations

(d) $a \to a_c$
fast fracture

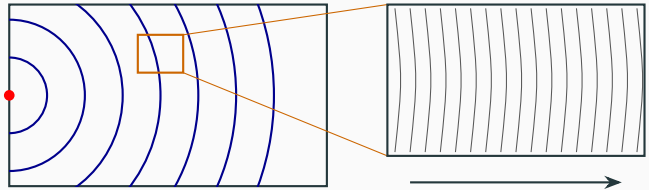
(a) and (b) are initiation — the S–N and ε–N methods of this lecture predict the cycles spent here. (c) and (d) are crack growth — fracture mechanics, next lecture. Note that nothing visible happens until late in the life.

Crack length against cycles: why inspection intervals exist



- Growth is accelerating: the longer the crack, the faster it grows
- Most of the life is spent at a size nobody can see
- The usable window runs from a_{det} (what NDT can find) to a_c (fast fracture)
- Inspections must fit at least twice inside that window

Reading the fracture surface: beach marks and striations

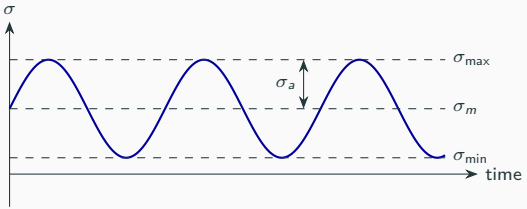


Beach marks are visible to the eye and mark changes of load *block* (a flight, a shutdown). Striations need an electron microscope and mark individual *cycles*. Counting back from the striation spacing is how an investigator reconstructs how long a crack had been growing.

Describing a cycle

The language of a cycle

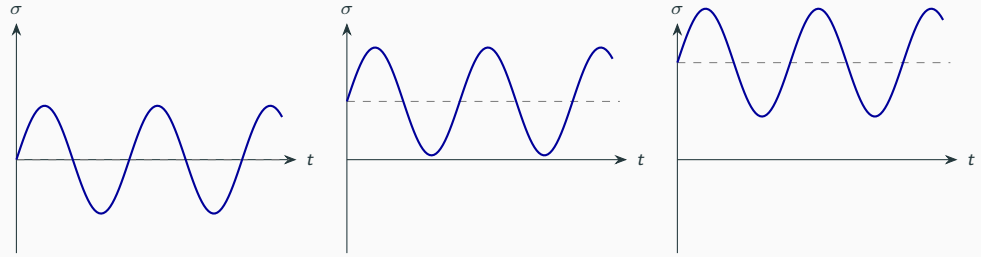
$$\sigma_m = \tfrac{1}{2}(\sigma_{\max} + \sigma_{\min})$$
$$\sigma_a = \tfrac{1}{2}(\sigma_{\max} - \sigma_{\min})$$
$$\Delta\sigma = 2\sigma_a, \qquad R = \frac{\sigma_{\min}}{\sigma_{\max}}$$



- $R = -1$ fully reversed — the reference for material data
- $R = 0$ zero-to-tension — pressurisation, landing
- $R = 0.1$ — the usual coupon test condition

Tensile mean stress is damaging; compressive mean stress is *beneficial* — which is why we shot-peen, cold-expand holes and use interference-fit fasteners.

The same amplitude, three different mean stresses

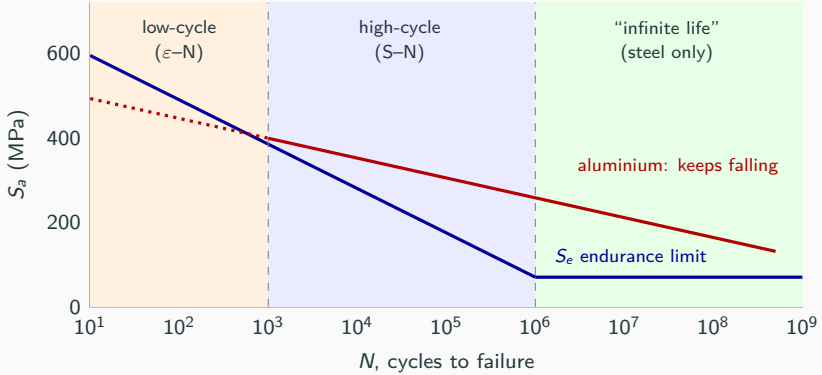


$R = -1$ fully reversed | $R = 0$ zero-to-tension | $R = 0.5$ high mean

All three have the same σ_a — and very different lives. Moving from $R = -1$ to $R = 0.5$ at constant amplitude can cost an order of magnitude in cycles. This is what the mean-stress corrections later in the lecture are for.

Stress-life

The S–N (Wöhler) curve and its three regions



- $N < 10^3$: plasticity dominates $\Rightarrow$ use ε–N
- $10^3 < N < 10^6$: straight on log–log, $S_a = aN^b$

- Steel and titanium: a true limit S_e at the knee — below it, infinite life
- Aluminium has no endurance limit: quote a strength *at a stated life* (5×10^8)

Classical correlations:

$$S'_e \approx 0.5\,S_{ut} \quad \text{(steel, } S_{ut} \le 1400\,\text{MPa)}$$

$$S'_e \approx 700\,\text{MPa} \quad \text{(steel above that)}$$

$$S_f@5\times10^8 \approx 0.4\,S_{ut} \ \text{(wrought Al)}$$

$$S_{1000} \approx 0.9\,S_{ut} \ \text{bending},\ 0.75\,S_{ut} \ \text{axial}$$

$$b = -\tfrac{1}{3}\log_{10}\frac{S_{1000}}{S_e}, \qquad a = \frac{S_{1000}}{(10^3)^b}$$

Correlations are a sanity check, not an allowable

For certification you use measured data: MMPDS Ch. 9 gives S–N curves per alloy, product form, K_t and R, fitted with the equivalent stress $S_{eq} = S_{\max}(1-R)^p$.

From specimen to component: the Marin factors

$$S_e = k_a\, k_b\, k_c\, k_d\, k_e\, k_f\, S'_e$$

k_a	surface	$a\, S_{ut}^{b}$: ground (1.58, -0.085), machined (4.51, -0.265), hot-rolled (57.7, -0.718), forged (272, -0.995). As-forged on 1200 MPa steel: $k_a \approx 0.23$
k_b	size	$1.24\, d^{-0.107}$ (d in mm); $k_b = 1$ for axial. Bigger part $\Rightarrow$ more initiation sites
k_c	loading	1 bending, 0.85 axial, 0.59 torsion
k_d	temperature	≈ 1 to 150 °C (Al), 450 °C (steel)
k_e	reliability	0.897 (90%), 0.868 (95%), 0.814 (99%), 0.753 (99.9%)
k_f	misc.	corrosion (removes the endurance limit entirely), fretting (3–10×), plating (-10 to -35%), residual stress

Fatigue starts at the surface — so the surface is where the designer's money goes.

Notches: K_t is not the whole story

$$K_f = 1 + q\,(K_t - 1), \qquad q = \frac{1}{1 + \sqrt{a}/\sqrt{r}}$$

- Static ductile design ignores K_t; fatigue cannot
- But the full K_t is not realised either: a sharp notch has a steep gradient, so little material sees the peak
- $\sqrt{a} \approx 0.25$–$0.5\ \sqrt{\text{mm}}$ (high-strength steel), $\approx 0.6\ \sqrt{\text{mm}}$ (aluminium)

The stronger the material, the more a notch punishes it

$q \to 1$ as strength rises. A 1800 MPa 4340 part with a sharp fillet can be worse in fatigue than a 900 MPa part with a generous one.

Apply K_f to σ_a (and conservatively to σ_m) — *or* divide S_e by K_f. Never both.

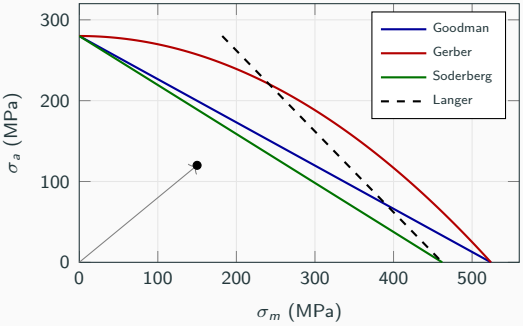
Mean stress: the constant-life diagram

$$\text{Goodman: } \frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{n}$$

$$\text{Gerber: } \frac{n\sigma_a}{S_e} + \left(\frac{n\sigma_m}{S_{ut}}\right)^2 = 1$$

$$\text{Soderberg: } \frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_y} = \frac{1}{n}$$

$$\text{Morrow: } \frac{\sigma_a}{S_e} + \frac{\sigma_m}{\sigma'_f} = \frac{1}{n}$$



- Goodman: simple, conservative, the design standard
- $\sigma_m < 0$: take $\sigma_a = S_e$, claim no credit
- Always add the Langer yield check

Variable amplitude: Palmgren–Miner

$$D = \sum_i \frac{n_i}{N_i}, \qquad \text{failure at } D = 1$$

What it assumes — and gets wrong

- Linear damage, no sequence effect
- Cycles below S_e ignored
- Test data fail anywhere from $D = 0.3$ to 3
- High-low is worse than low-high

Haibach: keep a shallower slope below the knee rather than truncating.

Counting the cycles

- Rainflow (ASTM E1049) — the only method consistent with the material's hysteresis memory
- Level-crossing and peak counting are cruder ancestors

Where the spectrum comes from

- GAG cycle (1/flight) — usually dominant
- Pressurisation (1/flight), gust and manoeuvre exceedance curves, taxi, landing, braking
- Ordered flight-by-flight for test and analysis

Worked examples

Example 2.1 — Machined 4340 shaft, 99% reliability

$S_{ut} = 1240\,\text{MPa}$, $d = 30\,\text{mm}$, machined, rotating bending

$$S'_e = 0.5 \times 1240 = 620\,\text{MPa}$$
$$k_a = 4.51 \times 1240^{-0.265} = 0.690, \qquad k_b = 1.24 \times 30^{-0.107} = 0.864$$
$$k_c = 1, \quad k_d = 1, \quad k_e = 0.814$$
$$S_e = 0.690 \times 0.864 \times 0.814 \times 620 = 297\,\text{MPa}$$

Add a fillet, $K_t = 1.8$, $q = 0.85$: $K_f = 1.68$, so allowable $\sigma_a = 297/1.68 = 177\,\text{MPa}$.

48% before the notch, 14% of S_{ut} after it

A stronger steel would raise S_{ut} but barely raise this number — k_a falls and q rises with strength. A bigger fillet radius and a better finish would do far more.

Example 2.2 — The fuselage lap joint

2024-T3, $\sigma_\theta = 75\,\text{MPa}$ once per flight from zero $\Rightarrow \sigma_a = \sigma_m = 37.5\,\text{MPa}$, $R = 0$.
Rivet hole: $K_t = 3.0$, $q \approx 0.9 \Rightarrow K_f = 2.8$. Local $\sigma_a = \sigma_m = 105\,\text{MPa}$.

$$\sigma_{a,\text{allow}} = 140\left(1 - \frac{105}{434}\right) = 106\,\text{MPa} \qquad (\text{Goodman, } S_f@10^6 = 140\,\text{MPa})$$

- Applied 105 MPa sits on the 10^6-cycle line — no margin for scatter
- A 60 000-flight service goal would need a scatter factor of $\sim$17
- And this ignores secondary bending from the eccentric lap — which can double it

Hence the real answer: lower hoop stress, cold-expanded holes, bonded doublers — or an explicit damage-tolerance justification with inspections.

Example 2.3 — Miner on a three-level spectrum

Per flight: 1 GAG at 120 MPa, 10 manoeuvre at 80 MPa, 200 gust at 40 MPa;

$S_a = 1100 N^{-0.13}$

Level	σ_a (MPa)	n/flight	N	n/N
GAG	120	1	2.5×10^7	4.0×10^{-8} (67%)
Manoeuvre	80	10	5.7×10^8	1.8×10^{-8} (30%)
Gust	40	200	1.2×10^{11}	1.7×10^{-9} (3%)
			D/flight	5.9×10^{-8}

Life $= 1.7 \times 10^7$ flights; with a scatter factor of 4, 4.2×10^6 certified.

The pattern is general

The one big cycle does two-thirds of the damage; 200 small ones do 3%. Reduce the *peaks*. Truncating the low end barely matters — truncating the high end is a serious error.

Strain-life

Why S–N runs out

- At a notch root under a large load the local material yields even though the nominal stress is elastic
- Stress then no longer characterises the loading: the notch root is *strain-controlled* by the elastic material around it
- Needed for $N < 10^4$, for notches in ductile alloys, and for thermal or displacement-controlled loading
- It is also the physically correct description of crack *initiation*, and the basis of modern fatigue software

Cyclic stress–strain

Metals cyclically harden or soften to a stable hysteresis loop: $\frac{\Delta\varepsilon}{2} = \frac{\Delta\sigma}{2E} + \left(\frac{\Delta\sigma}{2K'}\right)^{1/n'}$, with $n' \approx 0.1$–0.2.

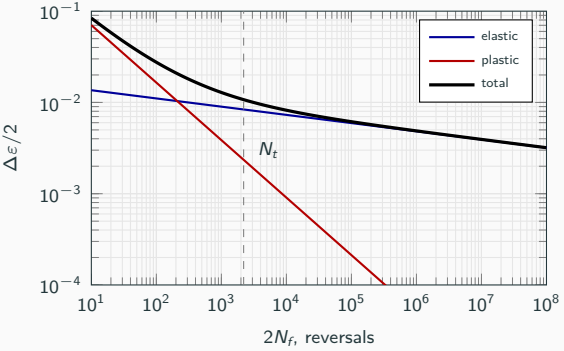
Coffin–Manson–Basquin

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma'_f}{E}(2N_f)^b + \varepsilon'_f(2N_f)^c$$

- $b \approx -0.05$ to -0.12 (strength)
- $c \approx -0.5$ to -0.7 (ductility)
- $2N_f$ = reversals, not cycles

The design dilemma, in one plot

Below N_t: ductility wins — a tough alloy outlives a strong one. Above N_t: strength wins.



Neuber's rule — getting local from nominal

$$K_f^2 = K_\sigma K_\varepsilon \quad \Longrightarrow \quad \sigma\varepsilon = \frac{(K_f S)^2}{E}$$

Solve this hyperbola *simultaneously* with the cyclic stress–strain curve; the resulting ε_a goes into Coffin–Manson.

Example 2.4 — 7075-T6, hole with $K_f = 2.6$, nominal $S_a = 150\,\text{MPa}$ (29% of S_{ut})

$$\sigma\varepsilon = (2.6 \times 150)^2/71\,700 = 2.12\,\text{MPa}$$

$$\Rightarrow\ \sigma \approx 379\,\text{MPa}, \quad \varepsilon \approx 5.59 \times 10^{-3}$$

$$\text{(elastic)}\ \ \sigma = 390,\ \ \varepsilon = 5.44 \times 10^{-3}$$

$$\Rightarrow\ 2N_f \approx 2.5 \times 10^5, \quad N_f \approx 1.2 \times 10^5$$

Two remarks

- The notch root yields on cycle 1 at less than a third of S_{ut} nominal — normal at a fastener hole
- This is life to initiation of a small crack. What remains is a crack-growth problem

Glinka's energy rule is less conservative; mean stress enters via Morrow or SWT.

Fatigue in the certification codes

CS-25.571: three philosophies

- Safe-life — retire the part at (demonstrated life)/(scatter factor $\approx$ 3–5). No cracks assumed. Landing gear, engine discs, some rotorcraft parts
- Fail-safe — a second load path carries limit load after any single failure. (The dual-lug fix from last lecture's error gallery)
- Damage tolerance — cracks are assumed to exist; inspections are scheduled to find them before they are critical. The default for transport metallic structure, and next lecture's subject

Where the scatter factor comes from

Every approximation in this lecture: S–N scatter, Miner's non-linearity, uncertainty in K_f, surface and environment, and the variability of the usage spectrum.

CS-25.571 and AMC 25.571 (EASA Easy Access Rules) · FAA AC 25.571-1D

Design levers, in order of effectiveness

1. Remove the stress concentration: larger radii, gentler transitions
2. Reduce the peak cycles — they dominate the damage sum
3. Improve the surface: finish, shot peening, cold-expanded holes, interference fasteners (cold expansion alone buys 3–5$\times$ life — the biggest single gain available)
4. Introduce compressive residual stress; avoid tensile mean stress
5. Protect from corrosion and fretting; specify plating and sealing
6. *Only then*: change the material or add thickness

Adding thickness is the last resort, not the first. It adds weight, and it does nothing about a sharp corner.

For next time

For next time

- Shigley Ch. 6, §§6.1–6.17; Dowling Ch. 9–10 and 14
- MMPDS Ch. 9: look up the S–N curves for 2024-T3 and 7075-T6 at $K_t = 1$ and $K_t = 3$ and compare
- Read CS-25.571 with its AMC
- Self-study problems 1–6 in the notes (two are Python: an ε–N solver and rainflow counting)
- Next: fracture mechanics — what happens *after* the crack exists, and how damage tolerance sets inspection intervals